INCH-POUND

MIL-B-85667/13
28 January 1991

MILITARY SPECIFICATION SHEET

BOLT, 100° TENSION FLUSH HEAD, STRAIGHT SHANK, ALUMINUM ALLOY SLEEVE, TITANIUM ALLOY PIN, 95 KSI Fsu, 160 KSI Ftu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-85667.

1. Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85729 shall be used with the sleeve bolts.

3. Oversize bolts. First and second oversize sleeve bolts may replace MIL-B-85667/13 sleeve bolts for first and second repair operations.

4. Installation. Installation shall be in accordance with MIL-HDBK-271-104.

5. Materials.

5.1 Pin. Pin material shall be titanium 6AL-4V in accordance with AMS 4967 or AMS 4928 (UNS R56400).

5.2 Sleeve. Sleeve material shall be in accordance with QQ-A-250/4 for aluminum alloys 2024-T3 (UNS A92024), 2024-T62 per QQ-A-250/4 (UNS A2024) and QQ-A-225/6 or QQ-A-200/3 for aluminum alloy 2024-T351 (UNS A92024).

6. Heat treatment.

6.1 Pin. The pin shall be heat treated in accordance with MIL-H-81200 and shall develop 160 KSI tensile strength and 95 KSI shear strength.

AMSC N/A FSC 5306

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

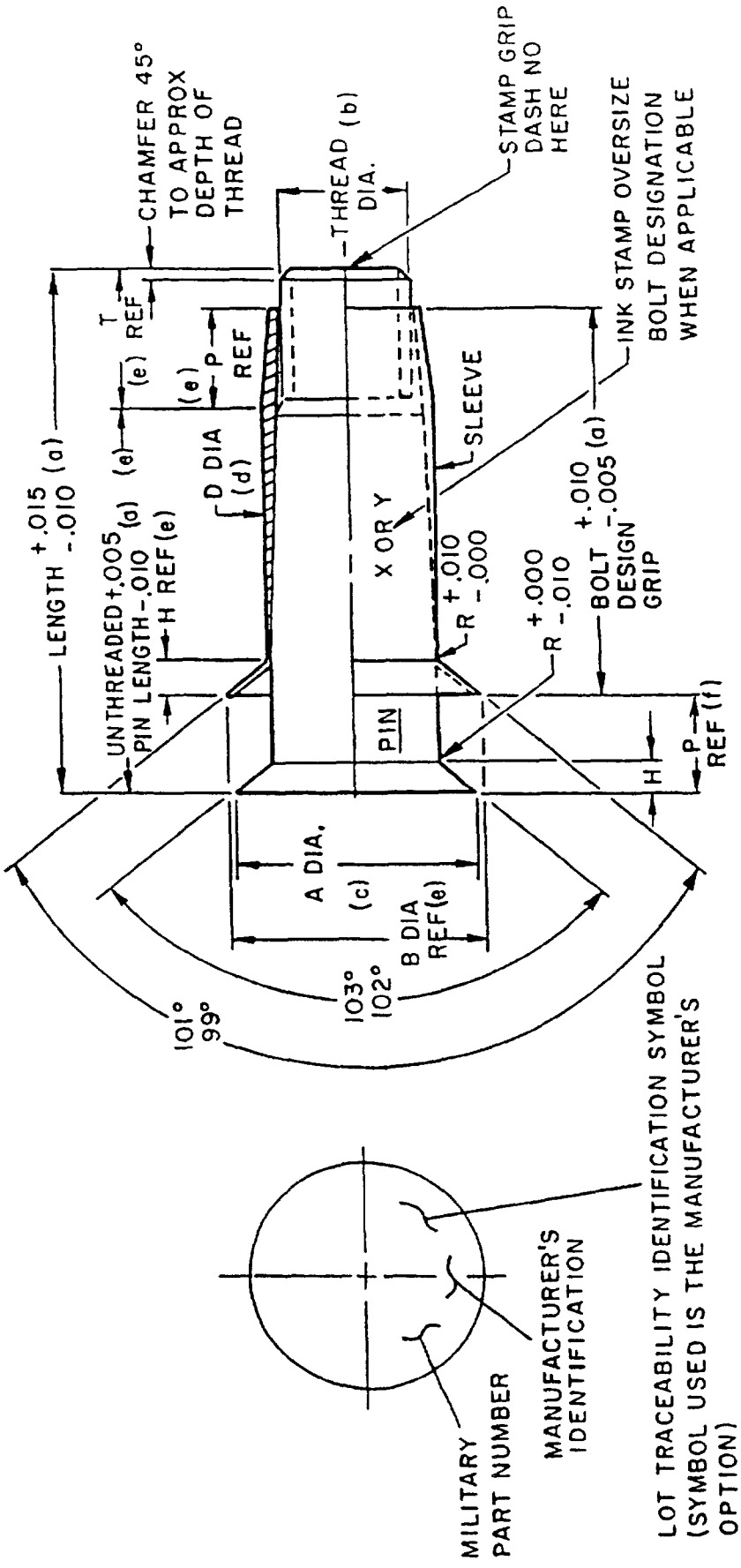



FIGURE 1. Tension flush head bolt.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

NOTES.

a. Grip lengths are available in .0625 increments.

b. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 FIM.

2. For sizes .437 thru .625, .012 FIM.

3. For size .750, .020 FIM.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 FIM.

2. For sizes .437 thru 0.750, .010 FIM.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. Protrusion will vary according to taper to provide expansion as given in Table I.

6.2 Sleeve. Aluminum alloy 2024-T62 sleeve shall be heat treated in accordance with QQ-A-250/5.

7. Surface texture. Surface texture shall be $R_a$-32 in accordance with ANSI/ASME B46.1.

8. Coatings or surface treatment.

8.1 Pin. The pin shall be aluminum coated in accordance with MIL-C-85614(AS).

8.2 Sleeve. The inside diameter of the sleeve shall be anodized in accordance with MIL-A-8625, Type I, Class 1. The outside diameter shall not be coated.

9. Lubricants. Dry film lubricant shall be applied to the pin outside diameter in accordance with MIL-L-46010, Type I, 0.0001-inch overspray allowed on head and threads. After application of dry lubrication, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE I. Bolt dimensions.

| Size Number | Group | Grip Dash Numbers | Thread Per MIL S 8879 UNJF 3A | A Bolt Head Dia | B Ref Sleeve Flange Diameter Std | B Ref Sleeve Flange Diameter X O/S | B Ref Sleeve Flange Diameter Y O/S | | Sleeve Flange Height Ref | R Ref | R Ref Radius | Thread A | Hex Angl | Dimension Initial Sleeve Diameter (A) Std | Dimension Initial Sleeve Diameter (A) X O/S | Dimension Initial Sleeve Diameter (A) Y O/S | F Dimension Final Sleeve Diameter (all) Std | F Dimension Final Sleeve Diameter (all) X O/S | F Dimension Final Sleeve Diameter (all) Y O/S | Minimum Installation Hole Size Diameter Std | Minimum Installation Hole Size Diameter X O/S | Minimum Installation Hole Size Diameter Y O/S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 7 thru 10 | 1900 32 | .416<br>.364 | .457 | .463 | .469 | .083 | .093 | .432 | .030 | .1640<br>.1610 | 343 | .2365<br>.2360 | .2425<br>.2420 | .2485<br>.2480 | .2455<br>.2445 | .2515<br>.2505 | .2575<br>.2565 | .237 | .243 | .249 |
| | 2 | 11 thru -16 | | .420<br>.370 | .459 | .465 | .471 | .081 | .093 | .432 | | | | .2365<br>.2360 | .2425<br>.2420 | .2485<br>.2480 | .2455<br>.2445 | .2515<br>.2505 | .2575<br>.2565 | .237 | .243 | .249 |
| | 3 | 17 thru -22 | | .430<br>.380 | .476 | .483 | .490 | | .095 | .432 | | | | .2485<br>.2480 | .2555<br>.2550 | .2625<br>.2620 | .2575<br>.2565 | .2645<br>.2635 | .2715<br>.2705 | .249 | .256 | .263 |
| 4 | 1 | 8 thru 10 | 2500 28 | .550<br>.494 | .586 | .597 | .603 | .110 | .120 | .2 | | | | .3005<br>.3000 | .3115<br>.3110 | .3175<br>.3170 | .3095<br>.3085 | .3205<br>.3195 | .3265<br>.3255 | .301 | .312 | .318 |
| | 2 | 11 thru -17 | | .553<br>.496 | .588 | .599 | .605 | .108 | .120 | .43 | .030 | .2440<br>.2410 | .406 | .3005<br>.3000 | .3115<br>.3110 | .3175<br>.3170 | .3095<br>.3085 | .3205<br>.3195 | .3265<br>.3255 | .301 | .312 | .318 |
| | 3 | -18 thru -28 | | .566<br>.515 | .607 | .614 | .623 | | .123 | .432 | | | | .3145<br>.3140 | .3215<br>.3210 | .3305<br>.3300 | .3235<br>.3225 | .3305<br>.3295 | .3395<br>.3385 | .315 | .322 | .331 |
| 5 | 1 | 8 thru 11 | 3125 24 | .684<br>.618 | .717 | .724 | .731 | .138 | .150 | .225 | | | | .3665<br>.3660 | .3735<br>.3730 | .3805<br>.3800 | .3755<br>.3745 | .3825<br>.3815 | .3895<br>.3885 | .367 | .374 | .381 |
| | 2 | 12 thru 20 | | .688<br>.622 | .719 | .726 | .733 | .136 | .148 | .432 | .040 | .3060<br>.3070 | .484 | .3665<br>.3660 | .3735<br>.3730 | .3805<br>.3800 | .3755<br>.3745 | .3825<br>.3815 | .3895<br>.3885 | .367 | .374 | .381 |
| | 3 | -21 thru -31 | | .702<br>.636 | .740 | .748 | .755 | | .151 | .432 | | | | .3805<br>.3800 | .3885<br>.3880 | .3955<br>.3950 | .3895<br>.3885 | .3975<br>.3965 | .4045<br>.4035 | .381 | .389 | .396 |
| 6 | 1 | -9 thru 11 | 3750-24 | .819<br>.746 | .848 | .856 | .868 | .166 | .175 | .225 | | | | .4315<br>.4310 | .4395<br>.4390 | .4515<br>.4510 | .4405<br>.4395 | .4485<br>.4475 | .4605<br>.4595 | .432 | .440 | .452 |
| | 2 | -12 thru -22 | | .823<br>.750 | .851 | .859 | .871 | .164 | .176 | .432 | .040 | .3680<br>.3640 | .562 | .4315<br>.4310 | .4395<br>.4390 | .4515<br>.4510 | .4405<br>.4395 | .4485<br>.4475 | .4605<br>.4595 | .432 | .440 | .452 |
| | 3 | -23 thru -34 | | .838<br>.765 | .872 | .880 | .888 | | .178 | .432 | | | | .4475<br>.4470 | .4555<br>.4550 | .4635<br>.4630 | .4565<br>.4555 | .4645<br>.4635 | .4725<br>.4715 | .448 | .456 | .464 |
| 7 | 1 | -9 thru -12 | 4375-20 | .956<br>.875 | .987 | .999 | 1.007 | .194 | .205 | .225 | | | | .4985<br>.4980 | .5105<br>.5100 | .5185<br>.5180 | .5075<br>.5065 | .5195<br>.5185 | .5275<br>.5265 | .499 | .511 | .519 |
| | 2 | -13 thru -26 | | .960<br>.878 | .991 | 1.003 | 1.011 | .192 | .207 | .432 | .050 | .4310<br>.4260 | .625 | .4985<br>.4980 | .5105<br>.5100 | .5185<br>.5180 | .5075<br>.5065 | .5195<br>.5185 | .5275<br>.5265 | .499 | .511 | .519 |
| | 3 | -27 thru -40 | | .978<br>.895 | 1.015 | 1.026 | 1.038 | | .208 | .432 | | | | .5185<br>.5180 | .5295<br>.5290 | .5415<br>.5410 | .5275<br>.5265 | .5385<br>.5375 | .5505<br>.5495 | .519 | .530 | .542 |
| 8 | 1 | -10 thru -12 | 5000-20 | 1.093<br>1.005 | 1.120 | 1.130 | 1.140 | .222 | .231 | .225 | | | | .5695<br>.5690 | .5795<br>.5790 | .5895<br>.5890 | .5785<br>.5775 | .5885<br>.5875 | .5985<br>.5975 | .570 | .580 | .590 |
| | 2 | -13 thru -30 | | 1.097<br>1.008 | 1.125 | 1.135 | 1.145 | .220 | .233 | .432 | .050 | .4930<br>.4880 | .703 | .5695<br>.5690 | .5795<br>.5790 | .5895<br>.5890 | .5785<br>.5775 | .5885<br>.5875 | .5985<br>.5975 | .570 | .580 | .590 |
| | 3 | -31 thru -46 | | 1.118<br>1.035 | 1.152 | 1.162 | 1.171 | | .236 | .432 | | | | .5895<br>.5890 | .5995<br>.5990 | .6085<br>.6080 | .5985<br>.5975 | .6085<br>.6075 | .6175<br>.6165 | .590 | .600 | .609 |
| 9 | 1 | -14 thru -23 | 5625 18 | 1.195<br>1.100 | 1.232 | 1.247 | 1.260 | .240 | .249 | .432 | .070 | .5550<br>.5500 | .781 | .6395<br>.6390 | .6545<br>.6540 | .6675<br>.6670 | .6485<br>.6475 | .6635<br>.6625 | .6765<br>.6755 | .640 | .655 | .668 |
| | 2 | 24 thru -48 | | 1.226<br>1.131 | 1.274 | 1.290 | 1.305 | .238 | .253 | .432 | | | | .6705<br>.6700 | .6865<br>.6860 | .7015<br>.7010 | .6795<br>.6785 | .6955<br>.6945 | .7105<br>.7095 | .671 | .687 | .702 |
| | 3 | -49 thru -72 | | 1.226<br>1.131 | 1.274 | 1.290 | 1.305 | | .253 | .900 | | | | .6705<br>.6700 | .6865<br>.6860 | .7015<br>.7010 | .6795<br>.6785 | .6955<br>.6945 | .7105<br>.7095 | .671 | .687 | .702 |
| 10 | 1 | -14 thru -25 | 6250 18 | 1.298<br>1.195 | 1.355 | 1.374 | 1.394 | .255 | .272 | .432 | .073 | .6160<br>.6120 | .843 | .7075<br>.7070 | .7265<br>.7260 | .7465<br>.7460 | .7165<br>.7155 | .7355<br>.7345 | .7555<br>.7545 | .708 | .727 | .747 |
| | 2 | -26 thru -52 | | 1.332<br>1.231 | 1.389 | 1.407 | 1.422 | .253 | .270 | .432 | | | | .7465<br>.7460 | .7645<br>.7640 | .7795<br>.7790 | .7555<br>.7545 | .7735<br>.7725 | .7885<br>.7875 | .747 | .765 | .780 |
| | 3 | -53 thru -78 | | 1.332<br>1.231 | 1.389 | 1.407 | 1.422 | | .270 | .900 | | | | .7465<br>.7460 | .7645<br>.7640 | .7795<br>.7790 | .7555<br>.7545 | .7735<br>.7725 | .7885<br>.7875 | .747 | .765 | .780 |
| 12 | 1 | -14 thru -29 | 7500-16 | 1.492<br>1.389 | 1.549 | 1.564 | 1.580 | .280 | .296 | .432 | .073 | .7430<br>.7370 | 1.000 | .8475<br>.8470 | .8575<br>.8570 | .8735<br>.8730 | .8515<br>.8505 | .8665<br>.8655 | .8825<br>.8815 | .843 | .858 | .874 |
| | 2 | -30 thru -60 | | 1.531<br>1.417 | 1.588 | 1.608 | 1.624 | .278 | .295 | .432 | | | | .8845<br>.8840 | .9045<br>.9040 | .9205<br>.9200 | .8935<br>.8925 | .9135<br>.9125 | .9295<br>.9285 | .885 | .905 | .921 |
| | 3 | -61 thru -90 | | 1.531<br>1.417 | 1.588 | 1.608 | 1.624 | | .295 | .900 | | | | .8845<br>.8840 | .9045<br>.9040 | .9205<br>.9200 | .8935<br>.8925 | .9135<br>.9125 | .9295<br>.9285 | .885 | .905 | .921 |

NOTES:

(A) D dia shall be verified within the grip bearing portion engaging the bolt shank prior to expansion; when bolt is installed to sleeve flange the dia. shall be within the expanded D dia. Limits shown in Table I.

(B) The sleevebolt is manufactured as a precision assembly and the components should never be dismantled. In the event of inadvertent separation, please discard and replace with a new assembly.

(C) All dimensions in inches, tolerances: decimals ±.010", angles ±2° unless otherwise specified.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE II. Mechanical properties.

| Dia Dash Number | Ultimate Tensile Strength Lbs-Min (b) | Thread Area $In^2$ (b) | Double Shear Strength Lbs-Min (a) | Pin Test Diameter (±.0005) In. (a) | Pin Test Dia Area $In.^2$ (a) | Max Install Torque In.-Lbs | Fatigue Test Loading Lbs ± 2% (c) | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Max | Min |
| -3 | 3,470 | .0217 | 5,380 | .1900 | .0284 | 60 | 1,600 | 160 |
| -4 | 6,210 | .0388 | 9,300 | .2500 | .0491 | 115 | 2,860 | 285 |
| -5 | 9,820 | .0614 | 14,600 | .3125 | .0767 | 185 | 4,520 | 450 |
| -6 | 15,200 | .0950 | 21,000 | .3750 | .1104 | 335 | 6,990 | 700 |
| -7 | 20,600 | .1288 | 28,600 | .4375 | .1503 | 540 | 9,480 | 950 |
| -8 | 27,500 | .1717 | 37,300 | .5000 | .1963 | 805 | 12,600 | 1,260 |
| -9 | 34,800 | .2176 | 47,200 | .5625 | .2485 | 1080 | 16,000 | 1,600 |
| -10 | 43,600 | .2724 | 58,300 | .6250 | .3068 | 1560 | 20,100 | 2,010 |
| -12 | 63,200 | .3952 | 83,900 | .7500 | .4418 | 2930 | 29,100 | 2,910 |

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 95 KSI and the stress area at the test diameter.

b. The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread and a tensile strength of 160 KSI.

c. Maximum fatigue load is based on 46% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

10. Example of military part number.

M85667/13 - 12 - 34

- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- '-' designates basic shank diameter. X designates 1st oversize, Y designates 2nd oversize. (Ink stamp on sleeve when applicable.)
- 12 — Diameter dash numbers in sixteenths of an inch.
- M85667/13 — Basic specification sheet number.

11. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE III. Bolt grips and lengths.

| Grip Number | Bolt Design Grip | -3 Diameter | | -4 Diameter | | -5 Diameter | |
|---|---|---|---|---|---|---|---|
| | | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 6 | .375 | .405 | .748 | .405 | .811 | .405 | .889 |
| 7 | .438 | .468 | .811 | .468 | .874 | .468 | .952 |
| 8 | .500 | .530 | .873 | .530 | .936 | .530 | 1.014 |
| 9 | .562 | .592 | .935 | .592 | .998 | .592 | 1.076 |
| 10 | .625 | .655 | .998 | .655 | 1.061 | .655 | 1.139 |
| 11 | .688 | .718 | 1.061 | .718 | 1.124 | .718 | 1.202 |
| 12 | .750 | .780 | 1.123 | .780 | 1.186 | .780 | 1.264 |
| 13 | .812 | .842 | 1.185 | .842 | 1.248 | .842 | 1.326 |
| 14 | .875 | .905 | 1.248 | .905 | 1.311 | .905 | 1.389 |
| 15 | .938 | .968 | 1.311 | .968 | 1.374 | .968 | 1.452 |
| 16 | 1.000 | 1.030 | 1.373 | 1.030 | 1.436 | 1.030 | 1.514 |
| 17 | 1.062 | 1.092 | 1.435 | 1.092 | 1.498 | 1.092 | 1.576 |
| 18 | 1.125 | 1.155 | 1.498 | 1.155 | 1.561 | 1.155 | 1.639 |
| 19 | 1.188 | 1.218 | 1.561 | 1.218 | 1.624 | 1.218 | 1.702 |
| 20 | 1.250 | 1.280 | 1.623 | 1.280 | 1.686 | 1.280 | 1.764 |
| 21 | 1.312 | 1.234 | 1.685 | 1.234 | 1.748 | 1.342 | 1.826 |
| 22 | 1.375 | 1.405 | 1.748 | 1.405 | 1.811 | 1.405 | 1.889 |
| 23 | 1.438 | 1.468 | 1.811 | 1.468 | 1.874 | 1.468 | 1.952 |
| 24 | 1.500 | 1.530 | 1.873 | 1.530 | 1.936 | 1.530 | 2.014 |
| 25 | 1.562 | | | 1.592 | 1.998 | 1.592 | 2.076 |
| 26 | 1.625 | | | 1.655 | 2.061 | 1.655 | 2.139 |
| 27 | 1.688 | | | 1.718 | 2.124 | 1.718 | 2.702 |
| 28 | 1.750 | | | 1.780 | 2.186 | 1.780 | 2.264 |
| 29 | 1.812 | | | | | 1.842 | 2.326 |
| 30 | 1.875 | | | | | 1.905 | 2.389 |
| 31 | 1.938 | | | | | 1.968 | 2.452 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE III. Bolt grips and lengths (continued).

| Grip Number | Bolt Design Grip | -6 Diameter | | -7 Diameter | | -8 Diameter | |
|---|---|---|---|---|---|---|---|
| | | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 6 | .375 | .405 | .968 | .405 | 1.030 | .405 | 1.108 |
| 7 | .438 | .468 | 1.030 | .468 | 1.093 | .468 | 1.171 |
| 8 | .500 | .530 | 1.092 | .530 | 1.155 | .530 | 1.233 |
| 9 | .562 | .592 | 1.154 | .592 | 1.217 | .592 | 1.295 |
| 10 | .625 | .655 | 1.217 | .655 | 1.280 | .655 | 1.358 |
| 11 | .688 | .718 | 1.280 | .718 | 1.343 | .718 | 1.421 |
| 12 | .750 | .780 | 1.342 | .780 | 1.405 | .780 | 1.483 |
| 13 | .812 | .842 | 1.404 | .842 | 1.467 | .842 | 1.545 |
| 14 | .875 | .905 | 1.467 | .905 | 1.530 | .905 | 1.608 |
| 15 | .938 | .968 | 1.530 | .968 | 1.593 | .968 | 1.671 |
| 16 | 1.000 | 1.030 | 1.592 | 1.030 | 1.655 | 1.030 | 1.733 |
| 17 | 1.062 | 1.092 | 1.654 | 1.092 | 1.717 | 1.092 | 1.795 |
| 18 | 1.125 | 1.155 | 1.717 | 1.155 | 1.780 | 1.155 | 1.858 |
| 19 | 1.188 | 1.218 | 1.780 | 1.218 | 1.843 | 1.218 | 1.921 |
| 20 | 1.250 | 1.280 | 1.842 | 1.280 | 1.905 | 1.280 | 1.983 |
| 21 | 1.312 | 1.342 | 1.904 | 1.342 | 1.967 | 1.342 | 2.045 |
| 22 | 1.375 | 1.405 | 1.967 | 1.405 | 2.030 | 1.405 | 2.108 |
| 23 | 1.438 | 1.468 | 2.030 | 1.468 | 2.093 | 1.468 | 2.171 |
| 24 | 1.500 | 1.530 | 2.092 | 1.530 | 2.155 | 1.530 | 2.233 |
| 25 | 1.562 | 1.592 | 2.154 | 1.592 | 2.217 | 1.592 | 2.295 |
| 26 | 1.625 | 1.655 | 2.217 | 1.655 | 2.280 | 1.655 | 2.358 |
| 27 | 1.688 | 1.718 | 2.280 | 1.718 | 2.343 | 1.718 | 2.421 |
| 28 | 1.750 | 1.780 | 2.342 | 1.780 | 2.405 | 1.780 | 2.483 |
| 29 | 1.812 | 1.842 | 2.404 | 1.842 | 2.467 | 1.842 | 2.545 |
| 30 | 1.875 | 1.905 | 2.467 | 1.905 | 2.530 | 1.905 | 2.608 |
| 31 | 1.938 | 1.968 | 2.530 | 1.968 | 2.593 | 1.968 | 2.671 |
| 32 | 2.000 | 2.030 | 2.592 | 2.030 | 2.655 | 2.030 | 2.733 |
| 33 | 2.062 | 2.092 | 2.654 | 2.092 | 2.717 | 2.092 | 2.795 |
| 34 | 2.125 | 2.155 | 2.717 | 2.155 | 2.780 | 2.155 | 2.858 |
| 35 | 2.188 | | | 2.218 | 2.843 | 2.218 | 2.921 |
| 36 | 2.250 | | | 2.280 | 2.905 | 2.280 | 2.983 |
| 37 | 2.312 | | | 2.342 | 2.967 | 2.342 | 3.045 |
| 38 | 2.375 | | | 2.405 | 3.030 | 2.405 | 3.108 |
| 39 | 2.438 | | | 2.468 | 3.093 | 2.468 | 3.171 |
| 40 | 2.500 | | | 2.530 | 3.155 | 2.530 | 3.233 |
| 41 | 2.562 | | | | | 2.592 | 3.295 |
| 42 | 2.625 | | | | | 2.655 | 3.358 |
| 43 | 2.688 | | | | | 2.718 | 3.421 |
| 44 | 2.750 | | | | | 2.780 | 3.483 |
| 45 | 2.812 | | | | | 2.842 | 3.545 |
| 46 | 2.875 | | | | | 2.905 | 3.608 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE III. Bolt grips and lengths (continued).

| Grip Number | -9 Diameter | | -10 Diameter | | -12 Diameter | |
|---|---|---|---|---|---|---|
| | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 14 | .905 | 1.686 | .905 | 1.748 | .905 | 1.905 |
| 15 | .968 | 1.748 | .968 | 1.811 | .968 | 1.968 |
| 16 | 1.030 | 1.811 | 1.030 | 1.873 | 1.030 | 2.030 |
| 17 | 1.092 | 1.874 | 1.092 | 1.935 | 1.092 | 2.092 |
| 18 | 1.155 | 1.936 | 1.155 | 1.998 | 1.155 | 2.155 |
| 19 | 1.218 | 1.998 | 1.218 | 2.061 | 1.218 | 2.218 |
| 20 | 1.280 | 2.061 | 1.280 | 2.123 | 1.280 | 2.280 |
| 21 | 1.342 | 2.124 | 1.342 | 2.185 | 1.342 | 2.342 |
| 22 | 1.405 | 2.186 | 1.405 | 2.248 | 1.405 | 2.405 |
| 23 | 1.468 | 2.248 | 1.468 | 2.311 | 1.468 | 2.468 |
| 24 | 1.530 | 2.311 | 1.530 | 2.373 | 1.530 | 2.530 |
| 25 | 1.592 | 2.374 | 1.592 | 2.435 | 1.592 | 2.592 |
| 26 | 1.655 | 2.436 | 1.655 | 2.498 | 1.655 | 2.655 |
| 27 | 1.718 | 2.498 | 1.718 | 2.561 | 1.718 | 2.718 |
| 28 | 1.780 | 2.561 | 1.780 | 2.623 | 1.780 | 2.780 |
| 29 | 1.842 | 2.624 | 1.842 | 2.685 | 1.842 | 2.842 |
| 30 | 1.905 | 2.686 | 1.905 | 2.748 | 1.905 | 2.905 |
| 31 | 1.968 | 2.748 | 1.968 | 2.811 | 1.968 | 2.968 |
| 32 | 2.030 | 2.811 | 2.030 | 2.873 | 2.030 | 3.030 |
| 33 | 2.092 | 2.874 | 2.092 | 2.936 | 2.092 | 3.092 |
| 34 | 2.155 | 2.936 | 2.155 | 2.998 | 2.155 | 3.155 |
| 35 | 2.218 | 2.998 | 2.218 | 3.061 | 2.218 | 3.218 |
| 36 | 2.280 | 3.061 | 2.280 | 3.123 | 2.280 | 3.280 |
| 37 | 2.342 | 3.124 | 2.342 | 3.185 | 2.342 | 3.342 |
| 38 | 2.405 | 3.186 | 2.405 | 3.248 | 2.405 | 3.405 |
| 39 | 2.468 | 3.248 | 2.468 | 3.311 | 2.468 | 3.468 |
| 40 | 2.530 | 3.311 | 2.530 | 3.373 | 2.530 | 3.630 |
| 41 | 2.592 | 3.374 | 2.592 | 3.435 | 2.592 | 3.592 |
| 42 | 2.655 | 3.436 | 2.655 | 3.498 | 2.655 | 3.655 |
| 43 | 2.718 | 3.498 | 2.718 | 3.561 | 2.718 | 3.718 |
| 44 | 2.780 | 3.561 | 2.780 | 3.623 | 2.780 | 3.780 |
| 45 | 2.842 | 3.624 | 2.842 | 3.685 | 2.842 | 3.842 |
| 46 | 2.905 | 3.686 | 2.905 | 3.748 | 2.905 | 3.905 |
| 47 | 2.968 | 3.748 | 2.968 | 3.811 | 2.968 | 3.968 |
| 48 | 3.030 | 3.811 | 3.030 | 3.873 | 3.030 | 4.030 |
| 49 | 3.092 | 3.874 | 3.092 | 3.935 | 3.092 | 4.092 |
| 50 | 3.155 | 3.896 | 3.155 | 3.998 | 3.155 | 4.155 |
| 51 | 3.218 | 3.998 | 3.218 | 4.061 | 3.218 | 4.218 |
| 52 | 3.280 | 4.061 | 3.280 | 4.123 | 3.280 | 4.280 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

TABLE III. Bolt grips and lengths (continued).

| Grip Number | -9 Diameter | | -10 Diameter | | -12 Diameter | |
|---|---|---|---|---|---|---|
| | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 53 | 3.342 | 4.124 | 3.342 | 4.185 | 3.342 | 4.342 |
| 54 | 3.405 | 4.186 | 3.405 | 4.248 | 3.405 | 4.405 |
| 55 | 3.468 | 4.248 | 3.468 | 4.311 | 3.468 | 4.468 |
| 56 | 3.530 | 4.311 | 3.530 | 4.373 | 3.530 | 4.350 |
| 57 | 3.592 | 4.374 | 3.592 | 4.435 | 3.592 | 4.592 |
| 58 | 3.655 | 4.436 | 3.655 | 4.498 | 3.655 | 4.655 |
| 59 | 3.718 | 4.499 | 3.718 | 4.561 | 3.718 | 4.718 |
| 60 | 3.780 | 4.561 | 3.780 | 4.623 | 3.780 | 4.780 |
| 61 | 3.843 | 4.624 | 3.843 | 4.889 | 3.843 | 4.843 |
| 62 | 3.905 | 4.686 | 3.905 | 4.748 | 3.905 | 4.905 |
| 63 | 3.968 | 4.749 | 3.968 | 4.811 | 3.968 | 4.968 |
| 64 | 4.030 | 4.811 | 4.030 | 4.873 | 4.030 | 5.030 |
| 65 | 4.093 | 4.874 | 4.093 | 4.936 | 4.093 | 5.093 |
| 66 | 4.155 | 4.936 | 4.155 | 4.998 | 4.155 | 5.155 |
| 67 | 4.218 | 4.999 | 4.218 | 5.061 | 4.218 | 5.218 |
| 68 | 4.280 | 5.061 | 4.280 | 5.123 | 4.280 | 5.280 |
| 69 | 4.343 | 5.124 | 4.343 | 5.186 | 4.343 | 5.343 |
| 70 | 4.405 | 5.186 | 4.405 | 5.248 | 4.405 | 5.405 |
| 71 | 4.468 | 5.249 | 4.468 | 5.311 | 4.468 | 5.468 |
| 72 | 4.530 | 5.311 | 4.530 | 5.373 | 4.530 | 5.530 |
| 73 | | | 4.593 | 5.434 | 4.593 | 5.593 |
| 74 | | | 4.655 | 5.498 | 4.655 | 5.655 |
| 75 | | | 4.718 | 5.561 | 4.718 | 5.718 |
| 76 | | | 4.780 | 5.623 | 4.780 | 5.780 |
| 77 | | | 4.843 | 5.686 | 4.843 | 5.843 |
| 78 | | | 4.905 | 5.748 | 4.905 | 5.905 |
| 79 | | | | | 4.968 | 5.968 |
| 80 | | | | | 5.030 | 6.030 |
| 81 | | | | | 5.093 | 6.093 |
| 82 | | | | | 5.155 | 6.155 |
| 83 | | | | | 5.218 | 6.218 |
| 84 | | | | | 5.280 | 6.280 |
| 85 | | | | | 5.343 | 6.343 |
| 86 | | | | | 5.405 | 6.405 |
| 87 | | | | | 5.468 | 6.468 |
| 88 | | | | | 5.530 | 6.530 |
| 89 | | | | | 5.593 | 6.593 |
| 90 | | | | | 5.655 | 6.655 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.

b. The bolt assemblies identified herein are proprietary products and are covered by the following patents U.S. 4,087,896 (expiring May 9, 1997), 4,106,036 (expiring July 25, 1997), 4,048,898 (expiring September 20, 1996), Great Britain 1,524,886; Canada 1,023,585, Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. The Government does not have a royalty-free license.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review:
Air Force - 82

Preparing Activity:
Navy - AS
Project No. 5306-1373



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:20Z
Check the source to verify that this is the current version before use.